Shape Matters: Galaxies Classification

Machine learning: Programming Assignment

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Abstract

The aim of the project is to build one or more classifiers that automatically identify the shape of a galaxy from its image.

1 Data Exploration and Preprocessing

The dataset comprises 11,000 RGB galaxy images, uniformly resized to 224×224 pixels and evenly distributed among four morphological classes. It is presplit into training (10,000 images) and test (1,000 images) sets, each organized by class. Given the balanced class distribution, accuracy serves as a valid and reliable performance metric.

A preliminary visual assessment reveals intra-class variations in brightness, likely attributable to differences in galactic distance—closer galaxies appear brighter. To address this, global brightness normalization is preferred over per-image scaling to retain class-defining features, such as the central luminosity gradient in elliptical galaxies. Per-image normalization risks attenuating these morphological cues.

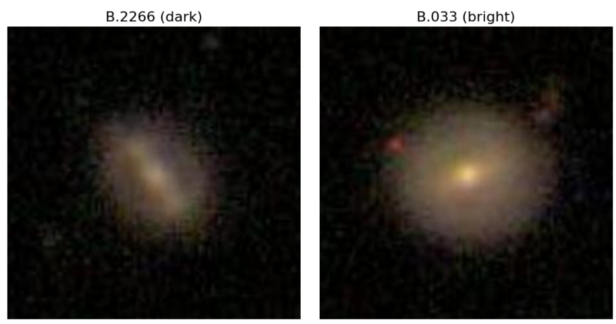


Figure 1: ***Two versions of barred galaxies — one brighter, one darker.***

Additionally, some galaxies within the same class exhibit noticeable color differences. For example, check out E-2151 and E-2203 in Figure (3). Although a general link exists between color and morphology (e.g., elliptical galaxies are typically redder, spirals bluer), color alone is not a dependable discriminator. This is evident from the overlapping class distributions in RGB median space (2), which lack clear separability. Therefore, for classification, all images were converted

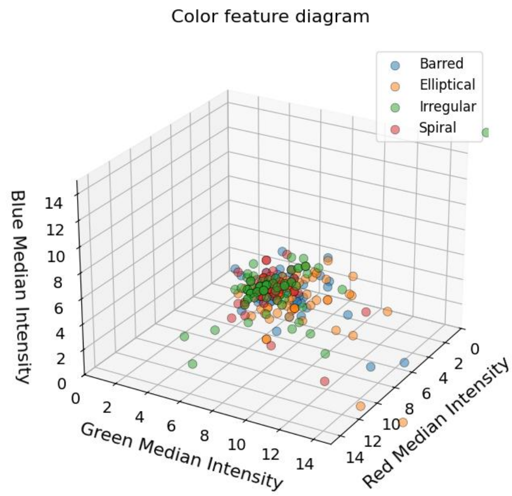


Figure 2: ***RGB-feature diagram:*** *shows the median color values of 70 different samples for each class.*

to grayscale and globally intensity-normalized (using mean-variance scaling). The role of color is further analyzed in Section 2.1. Finally, the defining traits of each class are summarized in Table (1).

2 A Classification Approach via Logistic Regression on Low-Level Features

Based on the observations summarized in Table (1), we build the following low level `feature_extractors` (number of features):

- `extract_symmetry_features` (3): quantifies radial and bilateral symmetry to aid in class differentiation.
- `cooccurrence_matrix` (64): encodes texture by measuring pixel-level co-occurrences.
- `edge_direction_histogram` (64): captures struc-

Figure 3: ***Galaxy sets, 5 from each class.*** *"C" stands for class, and the number points to the file in the folder.*

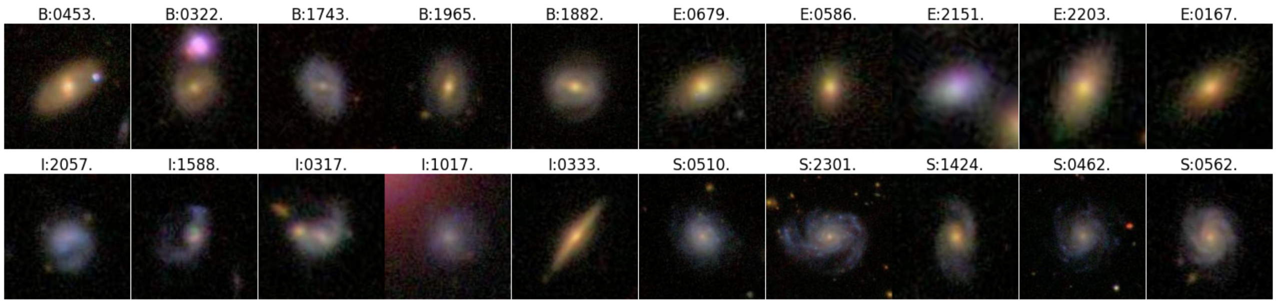


Table 1: ***Galaxy classification from visual observations.*** *Structural features (shape, symmetry) and secondary traits (color, brightness) per class.*

	Morphology			Secondary Obs	
Class	**Structure**	**Symmetry**	**Shape**	**Color**	**Brightness**
Spiral	*Spiral arms, central bulge*	*Axial*	*Flattened disk*	*Bluish arms, reddish bulge*	*High contrast*
Barred	*Linear bar, attached arms*	*Biaxial*	*Bar + disk*	*Reddish bar, bluish arms*	*Bar dominant*
Elliptical	*Smooth, featureless*	*Ellipsoidal*	*Rounded*	*Uniformly reddish*	*Uniform glow*
Irregular	*Clumpy, no structure*	*None*	*Amorphous*	*Patchy colors*	*Variable*

tural elements like spiral arms through edge orientation.

- `gini_batch` (1): computes the Gini coefficient to assess brightness concentration, useful for detecting core dominance.
- `grayscale_histogram` (64): summarizes global intensity distributions.

Some important points should be clarified: we applied global normalization (subtracting mean and dividing by std) to ensure features like texture matrices (large values) don't dominate symmetry scores (smaller scales) during training. This stabilizes gradient descent and balances feature contributions. While some features (e.g., co-occurrence matrices) are sparse, sparsity isn't explicitly handled because: (1) the linear model inherently ignores zeros, and (2) standardization implicitly reduces sparsity's impact. The chosen approach prioritizes simplicity, as the dataset is balanced and the logistic regression is robust to sparse, normalized inputs.
Let's proceed to extract some patterns.We do not expect brightness-related features to have a strong impact on morphology, but from a classification standpoint, they could be helpful. Although the accuracies are not particularly high, we can draw some interesting conclusions. The most notable is that, although the `All` configuration—combining all features—achieved the highest accuracy, the gap compared to `Shape_texture` (which excludes brightness-related features) is minimal (0.5%). This supports the idea that brightness features are not highly informative in morphological classification, confirming our initial idea. Another interesting aspect for a linear model is the slight underfitting observed in the training accuracies. This may be due to the use of very generic features (edges, shapes) rather than more specific ones. Finer features could include the number of spiral arms or the presence/absence of central bars. Moreover, when features are highly correlated, such as a color histogram and a texture descriptor, the model may receive overlapping information. For instance, if both features encode the presence of bright blue regions (common in spiral galaxies), they might carry redundant signals. In a linear model, this redundancy can lead to underfitting, because the model tries to distribute the importance across both features rather than amplifying the signal from one. Moreover, regularization tends to shrink weights, and when two features are similar, their weights may be jointly re-

Name	**Features**	**Accuracy(%)**	
		Train	**Test**
`Core_morphology`	`gr+sy+gi`	51.15	54.70
`Shape_Texture`	`ed+cooc+sy`	57.95	61.20
`Light_profile`	`gr+ed+gi+sy`	56.67	57.60
`Morphology_full`	`gr+symm+gi+ed`	58.99	59.80
`All`		59.32	61.70

Table 2: ***Trend (over 15 epochs) of the configurations.***

Figure 4: *Examples of misclassifications for the barred class.* `Shape_texture` *configuration.*

duced—even if the information is actually useful. As a result, the model underestimates the relevance of this characteristic and struggles to fit the data properly. This effect is less problematic in non-linear models like CNNs, which can learn interactions more flexibly, but it's a common limitation in feature-based linear classifiers.The CNN model will be discussed in detail in Section 3.
By experimenting with the `Shape_texture` configuration, we observe that it reaches an accuracy of 62, 20% after the 70 epoch. The most misclassified class is barred– as seen in Figure (5). According to the relative feature importance table (3), texture features dominate with 71.0%, followed by edge direction (18.7%)

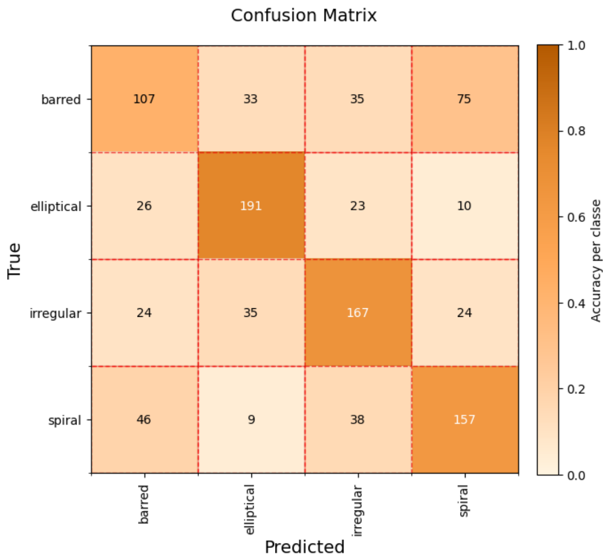


Figure 5: `Shape_texture` *confusion map.*

Feature Type	Importance (%)
`edge_direction_histogram`	18.7
`cooccurence_matrix`	71.0
`extract_symmetry_featuers`	10.3

Table 3: *Features' relative importance for* `Shape_texture` *configuration.*

and symmetry (10.3%). Barred galaxies are defined by a central bar that disrupts radial symmetry, which symmetry features may not fully capture. Overlap between edge and symmetry features further reduces their effectiveness. As a result, the model often confuses barred with spiral galaxies, which have similar textures but differ mainly in structure—whether a central bar or spiral arms—making it challenging for the model to determine whether the observed variation in texture arises from a linear bar or curved spiral arms. Figure (4) reinforces our earlier analysis. The first example (barred → irregular, score: 7.32) demonstrates an extreme case where the model misclassifies with abnormally high confidence—likely mistaking a distorted or asymmetric barred structure for irregular morphology. The second example (barred → spiral, score: 1.84) shows lower confidence but highlights how faint bar features might be misinterpreted as spiral arms.

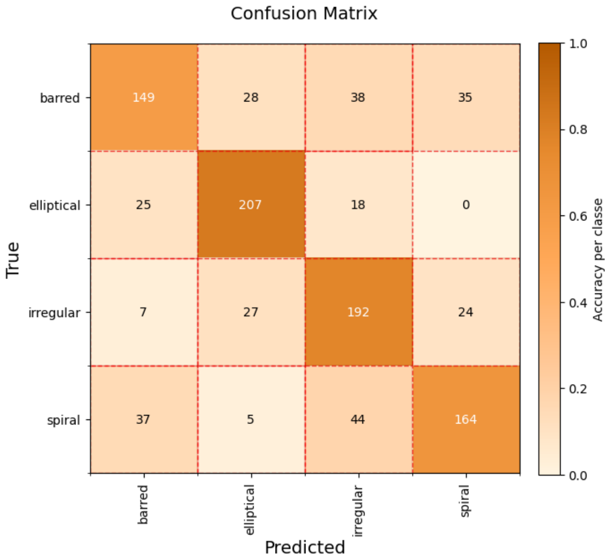
2.1 The Role of Color in Distinguishing Galaxy Types

In this configuration, RGB histograms are added to the `Shape_Texture` feature set (edge direction, RGB-based texture, symmetry). The model achieves a test accuracy of 71.20%, a clear improvement over the Shape_Texture baseline (61.20%). This may be explained by a partial complementarity between color histograms, which encode global chromatic distributions, and edge and texture features, which describe local morphology. The relative importance table (4)

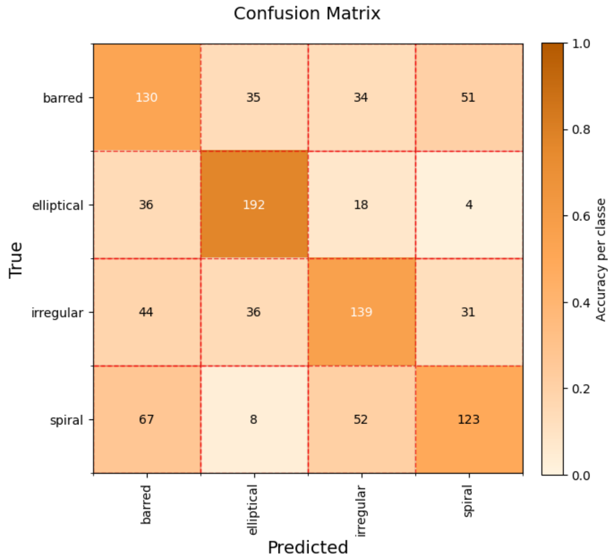
Feature Type	Importance (%)
`edge_direction_histogram`	13.7
`cooccurence_matrix`	58.9
`extract_symmetry_featuers`	7.3
`color_histogram`	20.1

Table 4: *Features' relative importance for* `RGB + Shape_texture` *configuration.*

shows that color histograms account for 20.1%, slightly more than any other group, but do not dominate the feature set. Texture features remain highly relevant (58.9%), edge direction follows (13.7%), and symmetry drops to 7.3%, suggesting some redundancy with color and local structure. The confusion matrix (6a) confirms that elliptical and spiral galaxies are well classified, likely due to their strong and distinct global and local patterns (e.g., color and arm structure). However, barred galaxies remain the most misclassified class, consistent with previous findings. This suggests that despite the addition of color, some structural cues—such as symmetry or bar orientation—may still be underrepresented or diluted when combined with stronger global features, due to the linear nature of the model.



(a) *Role of color:* `RGB + Shape_texture`.



(b) *Rotational invariance:* `RGB + Shape_texture`.

Figure 6: *Special request' confusion maps for the logistic regression classifier.*

2.2 Exploiting Orientation-Independence

The configuration `Texture+Edge` reaches an accuracy of 58.4% The removal of symmetry features—despite their modest 10.3% weight—degrades performance because they provide critical rotation-invariant cues. Barred and spiral galaxies exhibit distinct symmetry patterns (bilateral vs. radial) that persist across orientations, helping the model disambiguate them from irregular or elliptical types. Without symmetry, the model relies solely on edge and texture features, which are sensitive to rotation: a barred galaxy tilted at $\pi/4$ may appear smooth like an elliptical in edge maps, while its true bilateral symmetry would have corrected this misclassification. Similarly, faint spiral arms might be mistaken for asymmetric barred structures. Though low-dimensional, symmetry acts as a structural "tiebreaker," compensating for the limitations of rotation-variable features like edges. This explains why barred galaxies—whose classification hinges on recognizing their axis-aligned structure—suffer disproportionate misclassifications (particularly toward elliptical types) when symmetry is removed, even as other features dominate importance weights. The lesson is clear: for morphology tasks, even minor symmetry features provide essential robustness to rotational variance.

Figure 7: *Examples of misclassifications for the barred class.* `RGB + Shape_texture` *configuration.*

Feature Type	Importance (%)
`edge_direction_histogram`	14.1
`cooccurence_matrix`	85.9

Table 5: *Features' relative importance for* `Shape + texture` *configuration.*

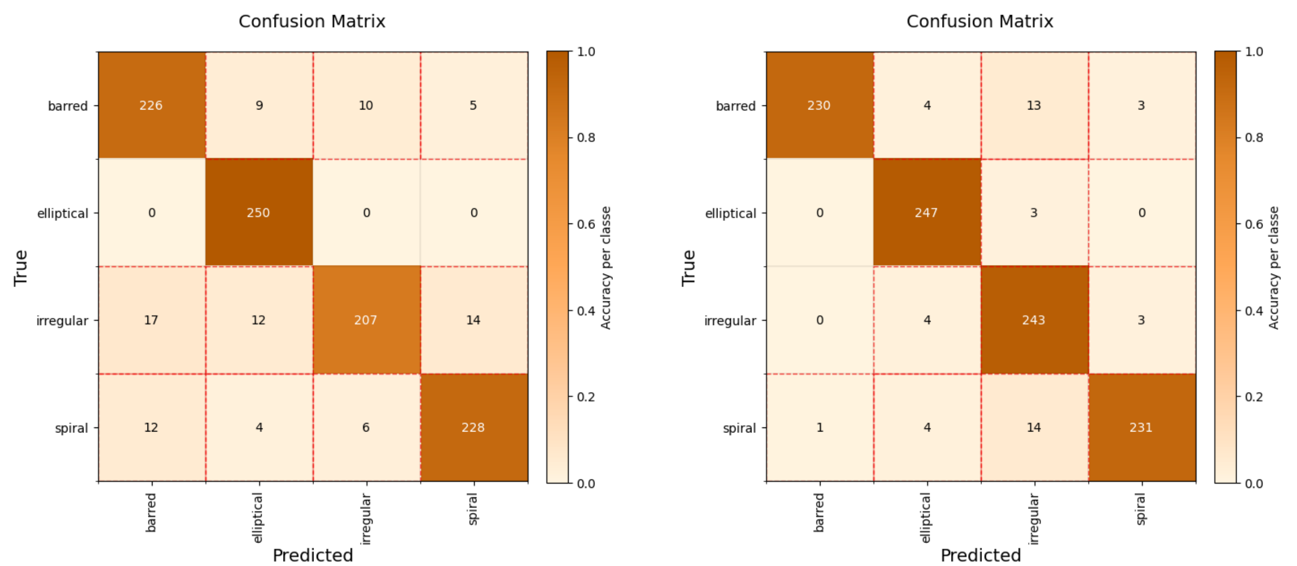
3 Deep Learning Approach: Transfer Learning and Layer-wise Fine-Tuning in ResNet18

This study employs a two-phase transfer learning approach using ResNet18 for galaxy classification. Initially, only the final fully connected layer is trained for 3 epochs while keeping all backbone layers frozen to leverage pretrained ImageNet features. Subsequently, a 17-epoch fine-tuning phase selectively unfreezes layers 3 and 4 (with reduced learning rates of 0.0001 vs. 0.001 for the classifier) to adapt mid/high-level features.

We tested several configurations: the baseline[1], the addition of color features, and augmentations compared to the baseline. The results are shown in Table (6).

Let's make a few key observations. Removing color information results in a modest decline in accuracy, indicating that color contributes to distinguishing certain

[1] CNN on the normalized image set as described in Section 1.



(a) *Baseline+RGB.* **(b)** *Baseline+Augmentation.*

Figure 8: *Special request' confusion maps for the CNN classifier.*

Configuration	Accuracy(%)	
	Train	**Test**
Baseline	99.12	89.60
Baseline + RGB	99.04	91.10
Baseline + Augmentation	95.32	95.10

Table 6: *CNN's results.*

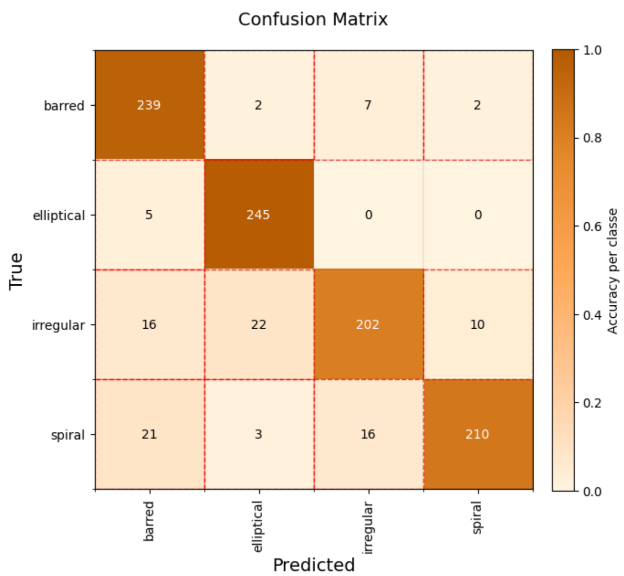


Figure 9: *Baseline's confusion matrix.*

galaxy types—particularly spirals, which often exhibit distinctive blue hues. Nevertheless, the model maintains strong performance, demonstrating that structural and luminosity-based cues alone encapsulate significant morphological information. Introducing rotational augmentations led to a significant boost in performance, especially for irregular galaxies, highlighting the value of rotation-invariant training in astronomy, where galaxies can appear in any orientation. This improvement is likely because irregular galaxies don't have a well-defined structure—their appearance stays roughly the same when rotated. They also don't rely heavily on color or fine details to be identified, making it easier for the model to recognize them based on their visual "disorder" alone.

To assess whether the model relies more on structural features—such as the number of arms,shape—rather than fine-grained cues, we applied a cartoon-like transformation that enhances edges while removing subtle textures. Fine-grained information may include internal texture, brightness gradients, or noise patterns. Despite their removal, the model maintained nearly 90% accuracy, suggesting a strong preference for global structural features over local details.

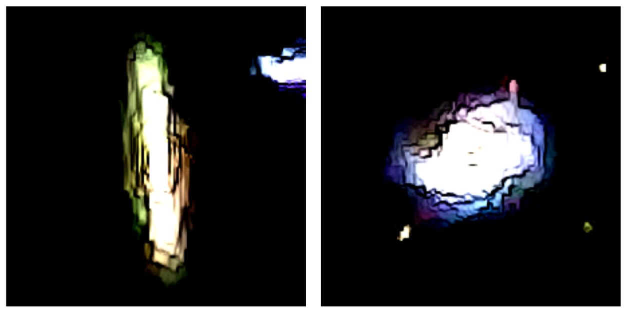


Figure 10: *Examples of* `cartoon` *galaxies.*

Declaration

I affirm that this report is the result of my own work and that I did not share any part of it with anyone else except the teacher.